COMSCI / ECON 206 | FINAL RESEARCH PROJECT | AUTUMN 2026

How to Make AI Applications Reduce Sycophancy

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Conditional model; one author reflection (n=1); purchases, live-LLM tests and learning outcomes remain unobserved.

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KEY MESSAGE

Paid correction needs reliable acceptance and enough net revenue.
Basic access also needs a funded audit and appeal route.

01 Question and foundations

Research question & verified gap

Can paid, independently accepted correction make apps honest while preserving access?
Case: a proposed tutoring service with two competing apps and three learners. Demand and budgets are assumptions.

Strategic model and information

Apps choose H (correction attempt) or S simultaneously; payoffs, values and costs are known, concurrent actions unobserved. Each H commits one attempt before bids.
Nash equilibrium predicts supply. Expected service return is q·price−c. Each attempt costs c.

Three complementary lenses

Game theory predicts supply; social choice compares surplus/access; mechanism design tests payments and review.
Observed n=1: 102 is correct; A preferred; stated WTP 3 vs 0; purchases n=0. These are distinct outcomes; demand stays assumed.

Auction and allocation boundary

Top k sealed bids win at bid k+1; lower ID breaks ties. One round, no reserve; losers pay zero. Fixed supply, private unit demand and q independent of bids. Budgets, biased checking and strategic auditors need further study.

02 Model and verified evidence

[1] Miller et al.
Truthful reports

[2] Witkowski et al.
Costly effort

[3] Yuan & Zhang
Independent auditing

This application: capacity → auction / funded lottery → independent check → release or refund

Honesty conditions: auction $qv_3 - c > 2$ and $qv_2 - c > 3$; funded lottery $qt - c > 3$.

Baseline profit (A,B)

A / B	H	S
H	(3,3)	(1,5)
S	(5,1)	(4,4)

Strong demand: (5,4,3).
At q=1, H−S=1−c.
c=1: every independent mixed pair is an equilibrium.
Weak (3,2,1): SS for c≥0.

Case	q	c	Equilibria	Output / limit
No service	—	—	SS	No correction
Strong auction	1	0	HH	Revenue 6; W=9
Strong auction	1	1	All pure/mixed pairs	Supply not unique
Acceptance test	.7	0	HH, SS; mixed H=2/3	No service can return
Funded lottery t=4	1	0	HH	Value 8; each access 2/3

Computational checks: 14,406 bid checks; 8 independent Nashpy matrices; 121 mixed pairs; 24 q/c cases. These tests verify the model and settlement rules.

03 Practical impacts and roadmap

Practical Impacts · SDG 4.4 / 4.5

Proposed education-access grant; learners pay 0.
k=2,q=1,t=4,c=0: reserve 9.5 = transfers 8 + audits .5 + appeal reserve 1. Delivery resource cost kc=0 is separate.

Key confirms 102 → pay 3, provider net 2.
Disputed 103 → hold 3 → independent rejection → refund 3, net −1 (c=1).
Audit .25 / appeal .50 separately funded.

4.4: delayed digital fact-checking vs ordinary answers.
4.5: completed access by budget/disadvantage.
Model baseline: auction v_3 access 0; lottery each 2/3.

Proposed indicators, not measured SDG effects. Value ≠ income. Exclusion, biased checks and deliberate basic errors require funded entry, independent appeals and free error reporting. Academic replication, provider logs and public-budget audits have different standards.

Future Research Roadmap

Economics: do payments cover costs? Real transactions; economist. Criterion: net margin/reserve. Milestone: preregister 1 month after approval.

CS: can checks be trusted? Held-out keys/appeals; CS educator. Criterion: error rates/logs. Milestone: benchmark before recruitment.

Behavior: accuracy or tone? Factorial test + spending; behavioral scientist/FP2 if interested. Criterion: separate effects. Milestone: ethics review, pilot, delayed test.

Full acknowledgments

I thank Prof. Luyao Zhang for suggesting AI sycophancy and separating well-being from engagement; Prof. Kenneth S. (Ken) Rogerson for his privacy question; Han Zhang for suggesting an honesty benchmark; Feiyu Li for questioning market-demand claims; Qingyue Liu for reviewing reproducibility, costs and access; and Yitong Lin and Qingyue Liu for their auditing and peer-prediction references, Incentive Clarity Award and accuracy–tone test suggestion. Longfei Jing contributed a PS1 review and Session F discussion; Tian Liang reviewed healthy recommendation algorithms; Zhengjun He joined the Session F discussion. Tencent's Shanghai office and the Shanghai Science and Technology Museum informed the field-learning reflections. I also thank the COMSCI/ECON 206 classroom community for feedback. The analysis draws on the cited research, Nashpy, Innovation Studio and official UN materials. I acknowledge GPT-6 Astra and GPT-6.1 Sol for assistance with assignment interpretation, writing and revision, references, figures, LaTeX, the Hugging Face and Colab materials, poster, README files and code verification (Appendix A.1). I remain responsible for this work's originality, accuracy and integrity.

Feedback and revision

Prof. Zhang (Oct 14): separate reflection from purchases and specify acceptance, appeals and lottery funding. These revisions retain cost-one indifference and the value 9 versus 8 comparison. FP2's literature suggestions inform independent checks; its award motivates the accuracy–tone study.

AI use: GPT-6 Astra (September 27, 2026) and GPT-6.1 Sol (October 6, 19 and 21, 2026) assisted with poster generation. I remain responsible for its originality, accuracy and integrity.

References · contribution · open materials

[1] Miller, N., Resnick, P. & Zeckhauser, R. 2005. Eliciting Informative Feedback: The Peer-Prediction Method. Management Science 51(9), 1359–1373. doi:10.1287/mnsc.1050.0379
[2] Witkowski, J., Bachrach, Y., Key, P. & Parkes, D.C. 2013. Dwelling on the Negative: Incentivizing Effort in Peer Prediction. HCOMP 1(1), 190–197. doi:10.1609/hcomp.v1i1.13089
[3] Yuan, D. & Zhang, J. 2026. When Complaints Go Quiet: Provider Strategy and Regulatory Design of AI Sycophancy. SSRN. doi:10.2139/ssrn.7455658

Yongzhuo Wu chose AI sycophancy as the research topic and developed the H/S model and allocation comparison. Verification uses notebook checks and Studio comparisons; the October 5 reflection informs the behavioral interpretation.

GitHub: github.com/dku-comsci-econ206-Autumn2026/PS2-FP1-Yongzhuo-Wu
HF: huggingface.co/spaces/dku-comsci-econ206-2026/AI-Sycophancy
Notebook: Colab/Final.ipynb